Solar Energy 245 (2022) 254–264

Contents lists available at ScienceDirect

Solar Energy

journal homepage: www.elsevier.com/locate/solener

Hybrid cooling techniques to improve the performance of solar photovoltaic modules

Ahmed Fawzy Eid [a,d,1], Sang-ik Lee [b,1], Seong-Gu Hong [c], Won Choi [d,*]

[a] Agricultural Engineering Research Institute, Agriculture Research Center, Giza, Republic of Egypt
[b] Department of Landscape Architecture and Rural Systems Engineering, Research Institute of Agriculture and Life Sciences, College of Agriculture and Life Sciences, Seoul National University, Seoul, Republic of Korea
[c] Rural System Engineering, School of Social Safety System Engineering, Hankyong National University, Anseong, Republic of Korea
[d] Department of Landscape Architecture and Rural Systems Engineering, Research Institute of Agriculture and Life Sciences, Global Smart Farm Convergence Major, College of Agriculture and Life Sciences, Seoul National University, Seoul, Republic of Korea

ARTICLE INFO

Keywords:
FEM
Photovoltaic module
Tedlar–polyester–tedlar
Thermal system
Thermoelectric cooling

ABSTRACT

Photovoltaic (PV) module efficiency is often impeded by concomitant temperature elevations, which may lead to cell degeneration and decreased electrical efficiency. Cooling techniques are commonly used to avoid this problem; however, current techniques, such as water-sprayer and active cooling systems, still have limitations, including the scattering and reflection of photons. In this study, we present a new active dual-cooling system design using two thermoelectric cooling (TEC) units fixed to the back of a PV module to provide rear-end cooling, which were experimentally and numerically studied. A water-thin film cooling system (WFCS) was maintained at the top surface of the module using a small affixed 72-W DC pump, for improving efficiency. For testing, the system was installed on the roof of an agricultural building in South Korea, and its performance efficiency was compared with that of a reference cooling-free PV module. The results showed that the innovation led to a decrease in the module temperature to 21 °C, which led to an efficiency improvement of 11.23 % with an increase in the daily output energy of 8.3 %. Additionally, a numerical simulation model based on the Finite Element Method (FEM) was used to evaluate the temperature profile of the PV module owing to the cooling technique and temperature distribution on the module's top surface. The maximum temperature reductions were 17.27 and 12 °C in the experimental and simulation results, respectively.

1. Introduction

By 2030, the world's energy consumption is expected to increase by 33 %, owing to several factors, such as the growing industrial sector, which consumes approximately 35 % of the total global energy output (Hasanuzzaman et al., 2012). An annual increase of 2.2 % in greenhouse gas (GHG) emissions was observed between 2000 and 2010 (Kalaugher, 2014). Additionally, according to the Fifth Evaluation Report of the Intergovernmental Module on Climate Change (2013), a 40 % increase in the concentration of atmospheric CO_2 was observed compared to the pre-industrial era, which is mainly a result of energy production by fossil fuel combustion, leading to unfavorable climate change and global warming (Manning et al., 2019). A solar photovoltaic (PV) module and solar collector were used separately to generate electrical and thermal energy, respectively from solar energy. Owing to recent technological advancements in silicon solar cell manufacturing, the conversion of 21 % of the incident solar radiation on the solar PV module into electricity is possible, and the remaining incident radiation accumulates inside the PV module and causes an increase in its temperature. This unused solar energy released from the solar PV module as heat increases the ambient temperature by 0.024 °C annually, which contributes to global warming (Terashima et al., 2020). To overcome this problem, different cooling techniques can be used to remove heat from PV modules and use it as thermal energy in many applications. To manage the global ambient temperature rise, solar photovoltaic thermal (PVT) technology could provide an opportunity for the complete use of solar energy employment to satisfy the demands of electrical and thermal energy (Chandrasekar and Senthilkumar, 2021).

* Corresponding author.
E-mail address: fembem@snu.ac.kr (W. Choi).
[1] Co-first authors who have worked together on a publication and contributed equally.

https://doi.org/10.1016/j.solener.2022.09.026
Received 17 May 2022; Received in revised form 3 August 2022; Accepted 14 September 2022
Available online 22 September 2022
0038-092X/© 2022 International Solar Energy Society. Published by Elsevier Ltd. All rights reserved.

Currently, renewable energy resources account for only 13 % of the total power supply (Borges Neto et al., 2010), and the growth in renewable sources (e.g., solar and wind systems) is forecasted to continue, owing to the pressing global demand for clean, renewable, and sustainable energy that does not harm the environment. Our planet receives a large amount of sunlight each day, which is sufficient to supply energy to the world for the entire year. Solar energy, in one form or the other, is the main source of all renewable energy resources on Earth. It is clean, secure and unswerving. Their availability makes them a highly attractive source of electricity.

A photovoltaic (PV) module is an outstanding example of an environmentally friendly electricity-generating source, where photons from the sun strike the surface of the semiconductor layer in the PV module, leading to the generation of electricity. In PV modules, photons that undergo reflection and/or transmission are lost and cannot be used to generate power. Some incident photons are converted into electricity; however, most of the remaining photons are converted into heat. This increase in temperature leads to a reduction in the bandgap of semiconductors. Silicon-based solar cells have bandgap of 1.1 eV. Thus, only incident photons with wavelengths greater than 1.1 eV pass from the layer of the solar cells and accumulate in the tedlar–polyester–tedlar (TPT) layer, causing a temperature increase (Teo et al., 2012).

Solar cells are sensitive to temperature, and an increase in temperature results in reduced efficiency. The open-circuit voltage (Voc) is mostly affected by temperature changes; however, the short-circuit current (Isc) increases only slightly with temperature (Hasanuzzaman et al., 2016). The maximum power (Pmax) of the module decreases by approximately 0.29–0.41 % for every degree of increase in module temperature above 25 °C under standard test conditions (STCs). For silicon solar cells, Voc decreases by 2.2 mV/ °C while, Isc increases slightly by 0.06 %/°C.

There are many technical possibilities for merging PV cells and solar thermal collectors. Commercial PVT collectors are available in the market and can be categorized into liquid and air collectors, owing to their basic design and heat medium (Brahim and Jemni, 2017). Photovoltaic thermal modules, typically abbreviated as PVT modules and also known as hybrid solar collectors, hybrid photovoltaic thermal solar collectors, PV/T collectors, and solar cogeneration systems, are power generation technologies that convert solar irradiation into usable thermal and electrical energy. Different types of PVT technologies have been reviewed, classified, and summarized in terms of their designs and efficiencies, such as air, water, and nanofluids based PVT collectors, PVT integrated with phase change materials (PCM), building-integrated PVT (BIPVT) systems, PVT attached with heat pipes, covered or glazed PVT collectors, uncovered or unglazed PVT collectors, and concentrated PVT collectors (Kumar et al., 2015; Chandrasekar and Senthilkumar, 2021).

This study can be classified as a PVT system based on a TEC device. A TEC was used to cool the PV module from behind, and the heat released from the PV module can be collected and used as thermal energy in different applications, such as domestic hot water heating, space heating, swimming pools, heat pump sources, solar water desalination, and agroindustrial processes. To evaluate the performance of a PVT collector, the electrical output and useful thermal energy from the collector are significantly important parameters. Khandelwal et al. (2007) tested non-contact and contact PVT models, and the electrical and thermal efficiencies were found to be 30–35 and 8–9 %, respectively. The exergy and energy efficiencies of a hybrid PVT collector-based air were found to be 55–65 and 12–15 %, respectively (Joshi and Tiwari, 2007). In terms of passive cooling techniques, PVT collectors based on PCM are the most promising for mitigating the temperature load of PV modules. In a previous study, the thermal and electrical outputs of the PV-PCM module were evaluated using a simulation and compared with those of a conventional PV module under the same operating conditions. The results showed an increase of 9.0 and 5.5 % in the peak electric power and daily output energy, respectively (Aneli et al., 2021). A concentrated PVT collector can be used to increase the output power from the unit area of the PV cells, in addition to reducing the system size (Renno, 2014). The authors conducted an overall review of various cooling techniques for PV modules in addition to a cost analysis of the effects of PV cooling on greenhouse gas emission reduction (Ghadikolaei, 2021a). A hybrid nanofluid-based PVT collector achieved the highest performance and thermal efficiency compared with a general water-based PVT collector (Ghadikolaei, 2021b). To increase the heat transfer from the PV module layers to the heat medium, a hybrid nanofluid comprising two or more types of nanoparticles mixed with the main fluid can be used (Ghadikolaei et al., 2017). A hybrid nanofluid composed of 50 % water and 50 % ethylene glycol achieved the highest heat transfer performance, owing to the high thermal conductivity of the mixture of two nanoparticles, compared to that of a pure nanofluid (Ghadikolaei et al., 2018a). Ghadikolaei et al. (2018b) analyzed the heat transfer of hybrid nanoparticles based on carbon nanotubes (CNTs)-water, and found that single-wall CNTs achieved higher heat transfer compared to multiwall CNTs. In addition, hybrid ethylene glycol CNT-based nanofluids was analyzed (Ghadikolaei et al., 2018c).

A thermoelectric cooler (TEC) enhances solar cell efficiency by pumping excess heat using the Peltier effect. Peltier effect-based devices are typically utilized for temperature control in PV-based applications such as concentrated PV systems, rooftop PV systems, tracking PV systems, and building integrated PV systems (Enescu and Spertino, 2017). The energy required to power a thermoelectric cooler device for temperature control in PV cells is acceptable (Najafi and Woodbury, 2013). Kane et al. (2017) described a TEC device for PV modules at ambient temperature and a solar insolation range of 800–1000 W/m^2. The temperature of the PV module decreased by 6–26 %, and the electrical efficiency improved by 1–18 %. Additionally, the TEC-equipped PV module was more effective at high ambient temperatures. Moreover, the heat removed by the cooling process can be used in other applications such as water heating. They can also be used in refrigeration applications. In another study, a thermoelectric cooling device (TEC) was attached directly to the backside of the PV module, and the electrical efficiency and output power of the PV module were found to improve by 1.10 % and 1.02 %, respectively (Kotkondawar et al., 2021). Dimri et al. (2018) used a TEC devices with two designs: (a) partially covered with TEC and (b) completely covered with TEC; the results showed that the total electrical efficiency of Case (b) was higher than that of Case (a) by 4.46–6.23 %. Singh et al. (2022) developed a combined TEC-PCM system that was attached to a solar PV module to cool it down; they also simulated the system using the COMSOL Multiphysics software to select the best combination, and found that the TEC achieved the highest performance under the same conditions. Zubeer and Ali (2022) tested and simulated a water- and nonaqueous-based concentrated photovoltaic (CPV), compared with a conventional PV module, and the results showed that the average PV module temperatures were 36.50, 64.10, and 57.50 °C, respectively.

In our experiment, TEC units with small fans and a heat sink were added to the system to distribute the cooled air to the backside areas of the PV module. This design achieved the highest performance in the cooling system, and we could reduce the required number of TEC units. Our approach was better than directly attaching the TEC to the PV, as tested by Kotkondawar et al. (2021). In addition, Siecker et al. (2017) and Aly et al. (2016) reported a new cooling system based on forced fan air, designed to alleviate the effects of elevated temperatures and dust accumulation on PV modules. This study showed a 48.4 W power gain for the described system compared with a reference cooling-free system. Another system based on active cooling by forced air circulation into a group of tubes attached to the back of the PV module was reported by Teo et al. (2012), showing an efficiency improvement of 12–14 % compared to the reference cooling-free system.

The net output power of a PV module is also affected by several factors. Moshfegh et al. (2018) reported a reduction in power output at high ambient temperatures, whereas increased wind speed, fin area, and solar radiation intensity led to an increase in the PV module power

output (Moshfegh et al., 2018). Also, Rahman et al. (2015) reported the effect of dust accumulation on the front side of the PV module and found that output power decreased by 7.7 W due to dust accumulation. Furthermore, an increase in the power output of 2.94 W was observed for every 100 W/m^2 solar radiation increase; however, the increase in solar cell temperature of 4.93 °C and relative humidity of 20 % led to a reduction in the electrical efficiency and output power by 0.3 % and 3.19 W, respectively.

In this study, we aimed to enhance the efficiency of a PV module by decreasing its temperature. To achieve this, we conducted an outdoor experiment with novel cooling technologies for PV modules, such as active thermoelectric cooling and an active water-thin-film-based closed-circulation cycle. The finite element method was used to simulate and analyze the physics of heat transfer and air velocity. The remainder of the paper is structured as follows: Section 2 discusses the materials and methods used in the study; Section 3 presents the comparative analysis of the experimental results and the FEM simulation results; finally, Section 4 presents the conclusions of the study.

2. Materials and methods

2.1. General conditions of the experiment

On September 18th, 2019, the designed PV modules were installed on the top roof of an agricultural engineering building at the Hankyong National University in South Korea. The mean ambient temperature in the experimental area was 24.5 °C under clear skies. The test was conducted between 9:00 am and 3:30 pm with good solar radiation of ~ 530 W/m^2 in the morning, maximum solar irradiance of 820 W/m^2 midday, and 700 W/m^2 at 3:30 pm. Meteorological data, including the ambient temperature, wind speed, humidity, and solar radiation, were obtained from a meteorological station near the location of the experiment. The PV modules were fixed at a tilt angle of 25° towards the south. The ambient wind speed was in the range of 3–5.5 m/s, and a water-based (20 °C) closed-circulation cycle was used for cooling. An overview of the system is shown in Fig. 1.

2.2. PV solar module

The PV module converts sunlight into electricity through a solar cell (worker) layer. A PV solar module consists of a group of solar cells connected in series, typically consisting of 36, 72, or 144 half-cut cells. A traditional PV module comprises different sandwich-like layers consisting of different materials, as shown in Fig. 2. In this study, two 100-W

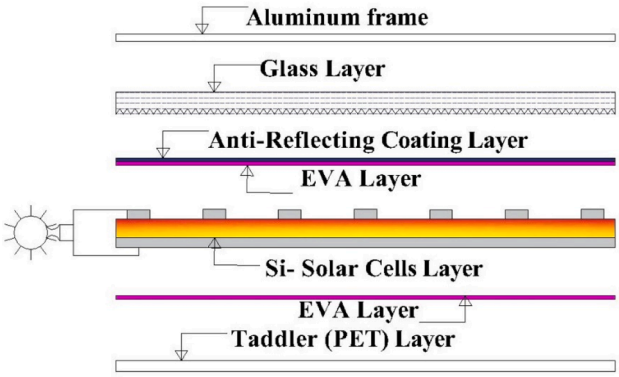


Fig. 2. Schematic diagram of the commercial PV module layers.

polycrystalline PV modules were used. One module was designed to have a dual cooling system (PV-WF/TEC, Model 1), and the other was employed as a reference with a non-cooling system (NCS, Model 2). The different components of the PV-WF/TEC system are listed in Table 1. An aluminum cover with a foam insulator was fabricated, as shown in Fig. 3.

2.3. TEC device

The active TEC device was composed of a Peltier effect-based small

Table 1
PV module characteristics and other PV-WF/TEC system components.

Parameter	Symbol	Value
Material	Si	Polycrystalline silicon
Maximum power (W)	P_{max}	100.00
Short circuit current (A)	I_{sc}	6.18
Open circuit voltage (V)	V_{oc}	21.60
Max. power current (A)	I_{max}	5.61
Max. power voltage (V)	V_{max}	17.80
PV Module dimensions (cm)	D	100.0 × 67.0 × 3.0
PV Cell dimensions (cm)	d	15.6 × 10.5
Number of cells	Nc	36 (4 × 9)
PV Module weight (kg)	W	8
Busbar width (mm)	B	1.5
Busbar length (mm)	L	104.0
Number of busbars	Nb	4
Pump Capacity (l/min)	Q	6
Thermoelectric cooler (mm)	D_TEC	40.0 × 40.0 × 3.8

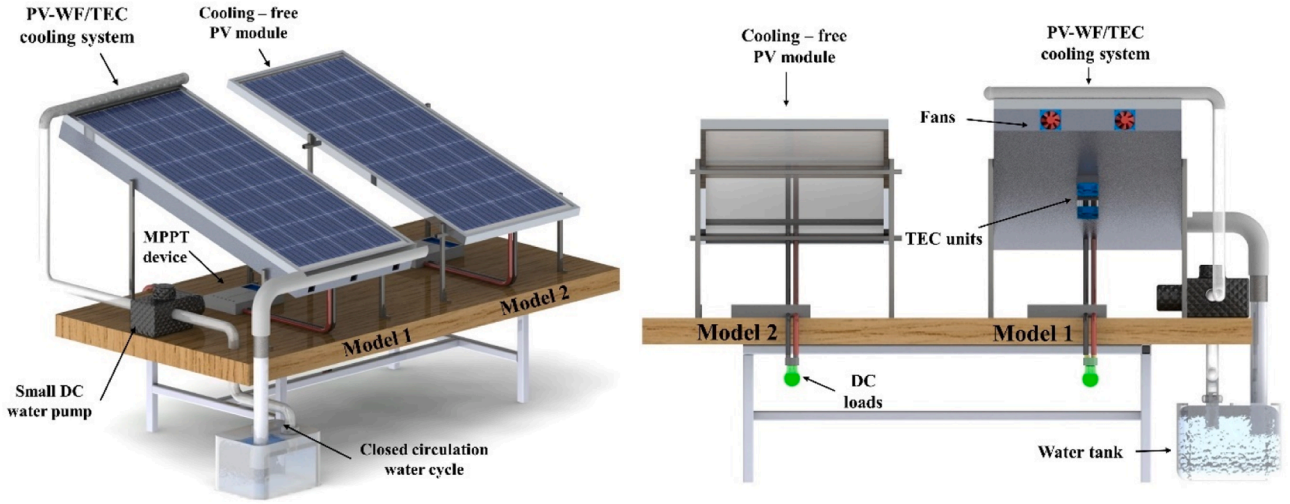


Fig. 1. An overall 3D view of the designed systems showing the PV-WF/TEC module (Model 1) and cooling-free module (Model 2).

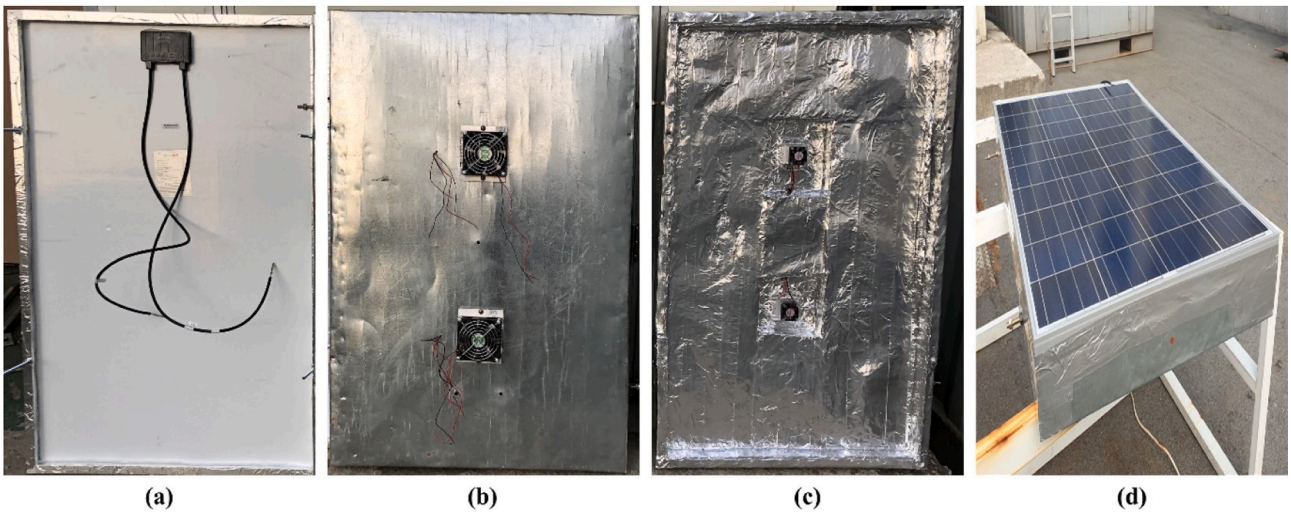


Fig. 3. Aluminum cover with a foam insulator fixed to the back of the PV module: (a) Back side of the PV module; (b) Back side of the aluminum cover; (c) Interior view of the aluminum cover; and (d) Overall view of the installation.

heat pump employing p- and *n*-type semiconductor pillars of Bi_2Te_3 with a low thermal conductivity of 1.5 W/m °C/m and a high electrical conductivity of 1.1×10^5 S/m for each square meter of unit, as shown in Fig. 4. The pillars were connected in parallel thermally and in series electrically using copper conductive tabs with a high thermal conductivity of 386 W/m °C, and were joined on both the upper and lower sides with ceramic plates of high thermal conductivity and high insulating capability. This TEC-based active cooling technique was used to cool the PV module from the backside.

An external DC supply was used to power the TEC unit and create a temperature difference between the upper and lower sides, rendering one side cold and the other hot. The TEC unit is designed to have two heat sinks and two fans per unit, as shown in Fig. 5. A pair of TEC cooler units was attached to the back of the aluminum cover, which was fixed to the back of the PV module. TEC devices have several advantages, as they do not require moving parts. Additionally, they have low maintenance, making them suitable for remote electricity provisioning.

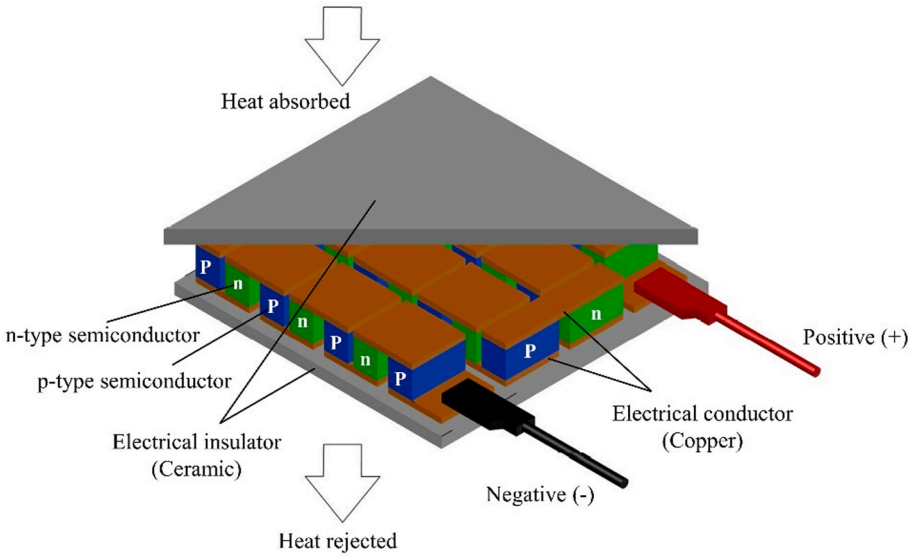


Fig. 4. Configuration of the thermoelectric cooler device.

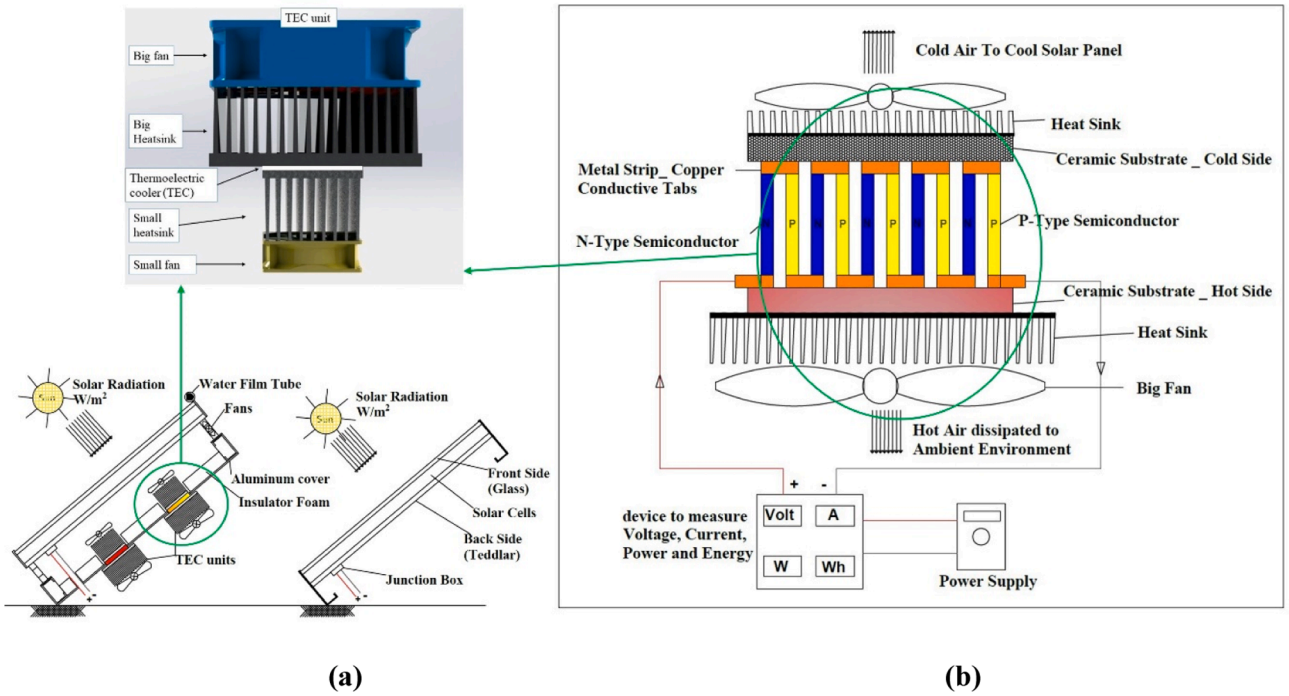


Fig. 5. Thermoelectric cooler device with heat sinks and fans: (a) 2D schematic of the workflow, and (b) Thermoelectric cooler device with heat sink and fans.

2.4. Measuring equipment and other components

A maximum power point tracker (MPPT) is a solar charge controller that constitutes a key component of the PV solar system. In addition, the MPPT can be defined as an electronic DC-to-DC converter that optimizes the match between the solar PV array and the battery bank or utility grid. That is, they convert a higher DC output voltage from the PV modules down to the required lower voltage for charging the batteries. The maximum power point (MPP) is the point of the highest efficiency on the solar module output curve, which can be attributed to the nonlinear output of the module. Conventional solar charge controllers are equipped with pulse-width modulation (PWM) charging technologies that cannot be used to track the MPP. However, in this study, solar charge controllers equipped with MPPT technology were used, as they could pick up the MPP of the module. The MPPT device was used instead of the PWM device, because it converts the excess input voltage into amperage, operates at Vmp rather than PWM, which operates below Vmp, provides more boost than PWM, especially during cold days and/or when the battery voltage is low, and is suitable for large module configurations that possess a lower cost per watt. In addition, high-efficiency MPPT solar-charge controllers (12 V DC, 10 A, 130 W; PTR1210AN, Photonic Universe, London, UK) were used. Monitoring station software was used to monitor output parameters and load consumption. The monitoring software was connected to the MPPT solar charge controller to track the power output from the developed solar modules and energy consumption by the loads. The LabVIEW® software was used to measure and record the temperature per second of the two PV modules on both sides using six K-type thermocouples. Each PV module had three K-type thermocouples, two on the back and one on the front.

A 72 W DC, 6 L/min pump (GOOD PUMPs, ROHS Motor, Beijing, China) was used to generate a thin film of water on the front side of the PV module to decrease the temperature, remove any accumulated dust on its surface, and improve efficiency. Two polyvinyl chloride (PVC) pipes were fixed on the upper and lower sides of the PV module. The upper PVC pipe delivered water, and the lower pipe discharged water back to the water tank, as shown in Fig. 1. A new method was used to mitigate the effects of dirt accumulation using a forced-air fan (Aly et al., 2016). Two small DC fans (DC 12 V, 0.2 A; Intel 1A21D7Q00, Intel, Shanghai, China) were installed on the top rear of the aluminum box to circulate air and release heat into the ambient air. A 12-V DC power supply (TDP-3030B, TOYOTECH, DongGuan, China) was used to power the water pump, TEC devices, and fans.

A small measuring device with a range of 0–20 A (DC 6.5–100 V) was used to measure the voltage, current, power, and energy consumed by the system. The net energy released by the cooling effect was calculated by subtracting the energy consumed by the loads and measuring devices, based on the energy generated from the PV modules. The measuring device was fixed between the power supply and TEC units to measure the aforementioned parameters. For the temperature measurements, six k-type thermocouples were attached at the front and back of the PV modules and connected to the LabVIEW®-based measuring device to record the temperature every second. For each model, two thermocouples were fixed at the backside of the PV module and the other was fixed at the front of the PV module. The thermocouple sensors were calibrated prior to the experiment with boiling water at 100 °C to confirm their accuracy. The tolerances for the thermocouples used in the experiment were within the acceptable range of ± 1 to 1.50 °C. In addition, an FLIR E8 WiFi thermal infrared camera (FLIR manufacture, Germany) with a thermal sensitivity of 0.10 °C, reading accuracy of ± 2.0 °C at ambient temperatures of 10–35 °C, and an IR resolution of 160 × 120 pixels, was used to analyze the temperature distribution across the front and back sides of the PV modules. The overall connections between the measuring devices, TEC units, power supply, and other components are shown in Fig. 6.

For efficiency calculation, the experimental conditions were set according to the datasheet of the PV manufacturer and standard test conditions (STCs) of 1,000 W/m^2 solar irradiance and 25 °C temperature. The maximum efficiency of the PV module was calculated as follows:

The efficiency was used to compare the PV module performance. The maximum efficiency is the ratio of the maximum output power (P_{max}) from the PV module to the input solar intensity power (P_{in}) from the sun, which is then converted to electricity. This can be determined by $\mu = (P_{max} / P_{in})$. The maximum PV module efficiency (17.8 %) at the STC was calculated from the maximum output from the PV module (100 W_p)

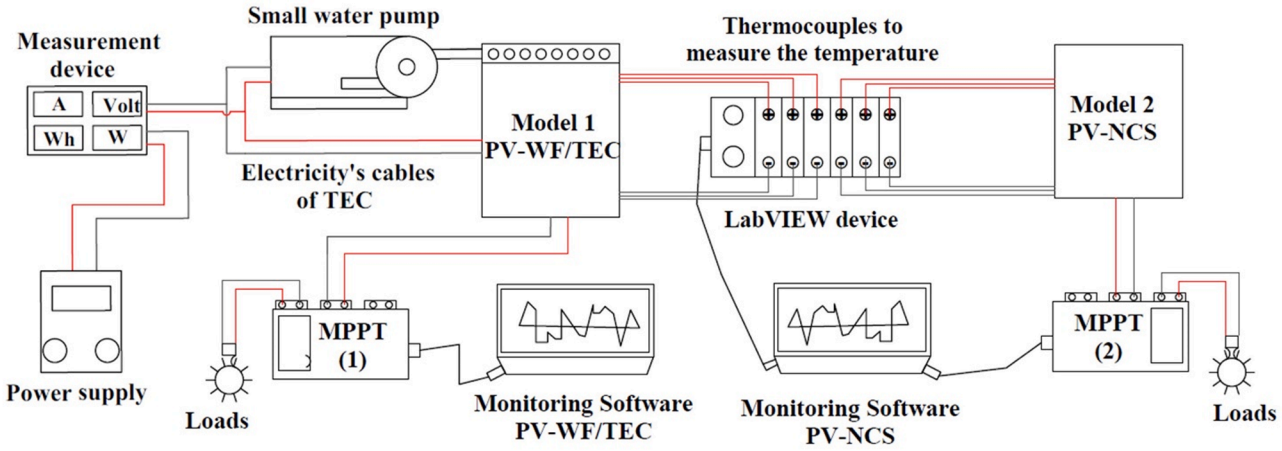


Fig. 6. Designed systems with measurement units.

divided by the solar radiation intensity of 1,000 W/m^2 multiplied by the net solar PV module area of 0.5616 m^2. The net solar PV module area was calculated as ((d × Nc) − (B × L × Nb)) / 10,000 × 0.5616 m^2. According to the thermal characteristics of the PV module, the PV efficiency decreased by 0.45 % for each degree of temperature above 25 °C. Therefore, the PV module efficiencies at different solar radiation and temperatures were calculated as 17.8 – (0.0045 × Δt × 17.8), where Δt is the temperature above 25 °C. However, real-world conditions usually differ from STC; therefore, a typical PV module seldom generates its calculated maximum wattage. The temperature of the PV module, for example, is seldom constant at 25 °C; most remain at approximately 30 °C, which is higher than the ambient temperature.

2.5. FEM simulation of the PV module

COMSOL® Multiphysics software applied FEM simulation to analyze the physics of heat transfer and air velocity in the studied solar module geometry. Different material properties were tested in each domain, as listed in Table 1. COMSOL® was used to divide the geometry into four domains, and the normal mesh-size technique was used to subdivide the main domains into a finite number of subdomains (elements), as shown in Fig. 7. The number of domain elements was 205042, and the skewness quality measurement yielded an average element quality was 0.5753.

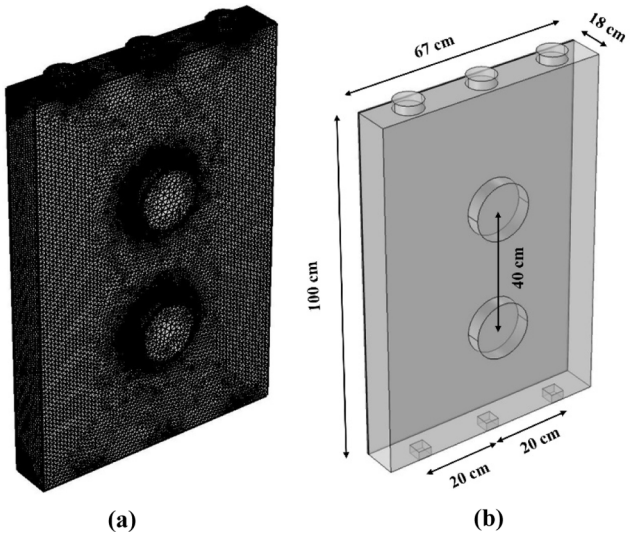


Fig. 7. 3D geometry of the PV-TEC model: (a) Normal mesh element size and (b) Transparent 3D geometry.

The material properties of the PV modules used in the simulations are presented in Table 2.

Some procedures for the development of new models for steady-state and unsteady calculations are time-consuming. There are certain numerical analyses and computational procedures that consume more than a month of CPU time. In our study, 16 cores with a base speed of 3.4 GHz and turbo were used to meet the recommended specifications for CFD. A multi-frontal massively parallel sparse (MUMPS) and generalized minimal residual (GMRES) solvers were applied for fluid dynamics and heat transfer, respectively, to minimize the computational time and to increase the convergence rate. A steady-state study was conducted to simulate the heat transfer across the PV module layer to analyze the temperature distribution and profile of the module under different ambient conditions. A time-dependent study was conducted to simulate the heat transfer and fluid flow across the PV-TEC-based modules as a function of time without a WF layer. Two geometries were designed using AutoCAD® software with a two-fan configuration. One applied the TPT layer domain, whereas the other did not. Three time points (0, 300, and 600 s) were tested for both geometries, with and without the TPT layer. The properties of the constituent domains of the geometries are presented in Table 2. Based on real-world data and uncontrolled ambient environmental conditions in the experimental zone, the average ambient conditions were set for the simulation. The average ambient temperature, atmospheric pressure, wind speed, and relative humidity were 27 °C, 1.01325 bar, 5 m/s, and 50 %, respectively. An average irradiance of 700 W/m^2 was applied under clear sky conditions.

3. Results and discussion

3.1. Experimental data

A significant change in temperature was observed in our proposed active dual-cooling PV-WF/TEC module, an average temperature decrease from 54 to 36 °C was observed at the top surface of the PV module, and a 3 °C decrease in temperature between the front and back sides of the PV-WF/TEC module was observed, compared with the PV-NCS module. Fig. 8 shows the effect of WF/TEC compared with that of NCS. The WF/TEC resulted in the lowest PV module temperature. The temperatures of the top surface of the PV module for the WF/TEC ranged from 28.76–to 36.08 °C, while the temperatures of the PV module for the NCS ranged from 45.82–to 52.87 °C. The maximum reduction in the average PV module temperature caused by using WF/TEC was 17.27 °C, whereas the difference in the temperature between the front and back sides of the PV module was in the range of 2–2.5 °C. The PV-WF/TEC model with a water-film-based closed circulation cycle on the glass surface resulted in good heat absorption, and the temperature of the

Table 2
Properties of the constituting layers of the PV solar module.

Material	Thermal Conductivity K, [W/m K]	Thickness t, [m]	Specific Heat Capacity Cp, [J/kg K]	Density ρ, [Kg/m^3]
Glass (quartz)	1.400	3.00×10^{-3}	730	2,210
EVA (Elvax 460 (18 % VA, 2.5 MI)	0.075	5.00×10^{-4}	2,301	1,200
Si - Polycrystalline Silicon	7.950	2.25×10^{-4}	690	2,800
PTFE - Polytetrafluoroethylene	0.414	1.00×10^{-4}	2,301	935
Aluminum	225.940	2.00×10^{-3}	921	2,698

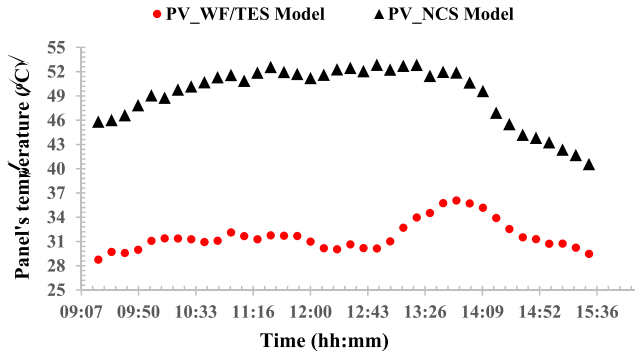


Fig. 8. Effect of the cooling system on the temperature of the PV module.

water increased from 23.9 to 35.9 °C, leading to an increase in the PV module temperature from 28.76 °C in the morning to 36.08 °C at 1:00 pm. This phenomenon was previously reported by Hosseini et al. (2011), who used a PVT system for heating water (Da Silva and Fernandes, 2010; Dannemand et al., 2019). This is because semiconductor devices such as solar cells are sensitive to temperature rise, which causes a reduction in the bandgap of the PV cell.

The PV module's efficiency was improved using the cooling system; thus, the PV-WF/TEC module achieved a higher performance than the PV-NCS module. The average hourly module efficiencies for PV-WF/TEC and PV-NCS were in the range of 17.08–17.44 % and 15.62–16.44 %, respectively, as shown in Fig. 9. For PV-NCS, an average temperature rise of 52.20 °C led to a decrease in efficiency by an average of 13.6 %. However, the average hourly PV module efficiency for PV-WF/TEC improved owing to the cooling effect of approximately 10.3 %. Additionally, owing to the increase in water temperature caused by the closed water circulation over time, the average hourly PV module efficiency of the PV-WF/TEC improved by 5.2 %. Harmonic results were discussed by Abdelrahman et al. (2013), and from this, the water-cooling method is suitable for cleaning the PV surface from accumulated dust. However, this can decrease the efficiency by decreasing the number of photons in the PV module (Bouraiou et al., 2015).

With respect to the cooling effect on the PV module temperature, the temperature of water in the closed-circulation cooling system increased over time because of heat absorption from the PV module, which resulted in a decrease in the electrical efficiency from 17.4 to 16.9 % as the temperature increased from 28 to 36 °C. The results also showed that the electrical efficiency decreased with increasing temperature; however, using the WF/TEC technique to decrease the temperature of the PV module led to an improvement in electrical efficiency, as shown in Fig. 10.

As shown in Fig. 11, the power density has a direct effect on the output power of the PV modules. The figure illustrates the increase in the output power as a function of solar radiation levels. However, more than 70 % of the power striking the PV module is converted to heat, leading to an increased temperature and a subsequent reduction in the output power. In this study, we observed that the use of cooling effects in PV-WF/TEC led to an increase in output power by 9.32 % at a maximum solar intensity value of 819.44 W/m^2 at 1:00 pm.

As shown in Fig. 12 (a), the highest reported ambient wind speed of 5.1 m/*sec* was observed at 2:00 pm, which led to a decrease in the PV module temperature and improved PV module performance. The effect of ambient temperature on the PV-NCS model was also monitored, as shown in Fig. 12 (b). An increase in the temperature midday resulted in a subsequent increase in the temperature of the PV module, leading to further performance deterioration. This harmful temperature effect was also previously reported by Razak et al. (2016).

As shown in Fig. 13 (a), the output voltages at the MPP from all the PV modules decreased with increasing temperature. The voltage reduction was a direct result of the reduction in the semiconductor bandgap inside the solar cells caused by the temperature increase and the subsequent elevation in the concentration of intrinsic carriers, thereby the decreasing voltage at maximum power (V_{mp}). The utilization of our proposed dual-cooling system resulted in an improved V_{mp}, when compared with the NCS PV module.

Fig. 13 (b) shows a slight increase in I_{mp} as a function of temperature, indicating that the changes in temperature experienced by the WF/TEC- and NCS-based PV modules did not have a considerable impact on I_{mp}. That is, I_{mp} values were nearly the same for both WF/TEC-and NCS-based PV modules, despite the variations in their experiences with temperature (Hindee et al., 2016).

Owing to the cooling effect, an increase in average V_{mp} from 19.31 to 20.53 V was experienced in the PV module when the average temperature decreased from 49.3 to 31.6 °C, respectively. The average I_{mp}

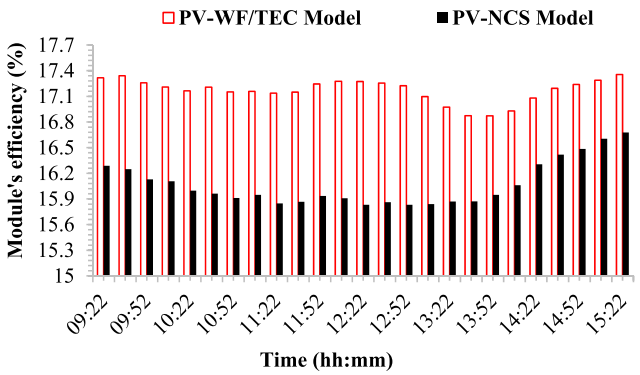


Fig. 9. Effect of cooling on the electrical efficiency of PV modules.

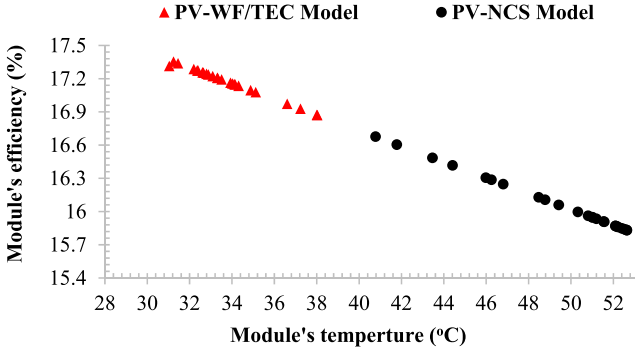


Fig. 10. Impact of temperature rise on the electrical efficiency of the PV modules.

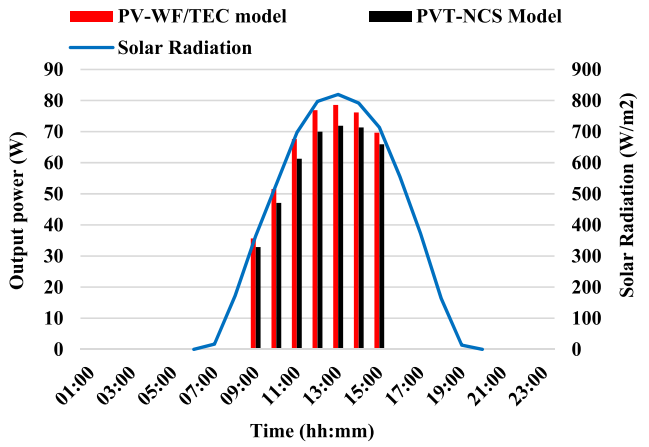


Fig. 11. Effect of cooling on the output power of the PV modules.

increased from 1.898 to 2.005 A when the average temperature of the PV module decreased from 49.3 to 31.6 °C, respectively.

Our results show that the daily output energy from the PV module increases with the cooling effect compared with the non-cooling-based modules. The daily output energy improved by 8.3 % when using WF/TEC, as shown in Fig. 14. During the experiment time, the energy generated from the PV modules using WF/TEC and NCS was 0.458 and 0.42 kWh, respectively, and the energy consumed by the cooling system was 0.076 kWh. The net energy generated by the PV module using the WF/TEC technique was 0.382 kWh.

Finally, despite the improvement in the output energy and power of the PV module owing to the cooling effect, the net generated energy was unsatisfactory owing to the increased energy consumption caused by the use of a high-capacity water pump and TEC units, the low thermal conductivity of the TPT layer on the back side of the PV module, and the aluminum-based backside cover, which led to increased heat accumulation.

3.2. Finite element simulation

Using a steady-state study, the temperature profile and distribution within the PV module layers were simulated using the COMSOL® Multiphysics software. The results showed a maximum temperature of 66 °C at the solar-cell layer in the cooling-free PV module, which was expected because the solar cell layer was the effective layer where radiation absorption occurred. Additionally, the resulting simulation predicted a relatively low temperature on the top surface of the glass layer compared to the back of the PV module, as shown in Fig. 15. In the case of steady-state analysis, the results showed that the simulation output was harmonic with the actual experimental results. Under the

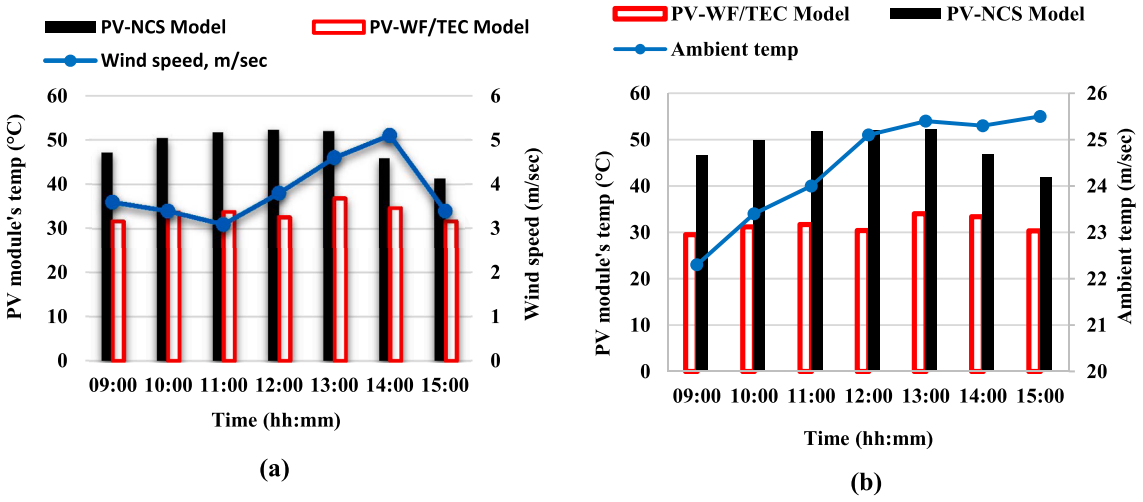


Fig. 12. Impact of ambient environment on PV module temperature: (a) Wind speed and (b) Ambient temperature.

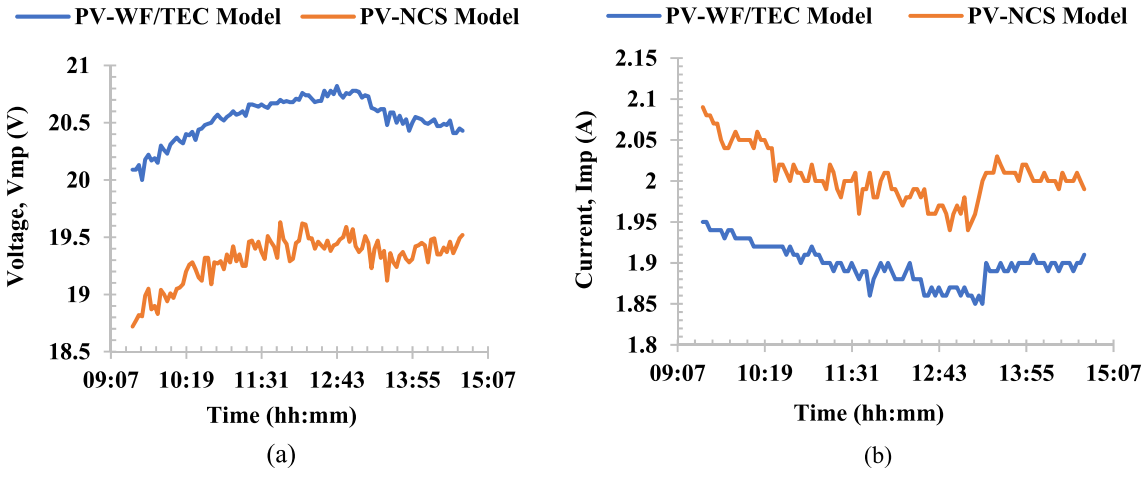


Fig. 13. Impact of cooling and non-cooling system on the PV modules outputs at maximum power.

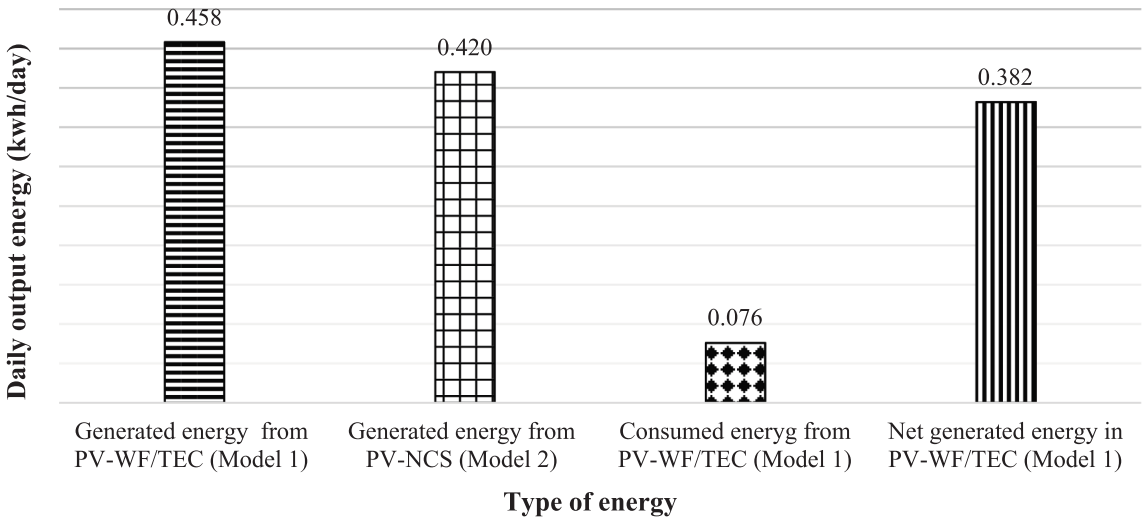


Fig. 14. Daily output energy obtained due to the cooling effect.

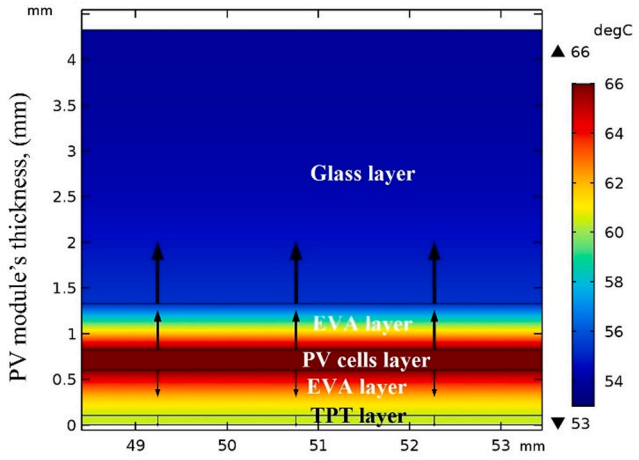


Fig. 15. Temperature distribution within the different layers of the cooling-free PV.

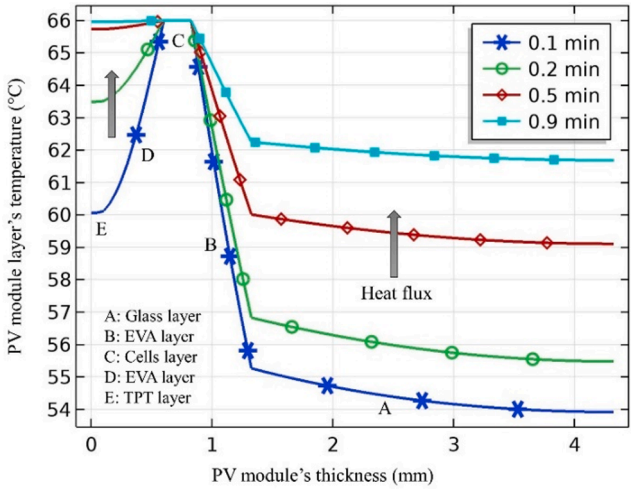


Fig. 16. Temperature profile of the cooling-free PV module layers.

same ambient conditions, the top surface temperatures of the PV (glass layer) were 54 and 52.87 °C for the simulation and experiment. In addition, the simulation results showed that there is a difference of temperature between the front and backside surfaces of the PV module, which is similar to the actual test. The simulation study provided finer details with respect to all PV layer temperatures, which are difficult to measure in a real experiment, such as those of the solar cell layer (active layer) and the EVA layer.

Fig. 16 shows the temperature profiles of the different layers with the NCS in a time-dependent manner. At a time of 0.1 min, a maximum temperature of 66 °C was observed in the solar-cell layer, whereas a minimum temperature of 54 °C was observed on the top surface of the glass layer. A difference of approximately 8 °C was observed between the front and back of the PV module. The results of the time-dependent simulation were in agreement with the experimental results. Owing to the different thermal conductivities of the glass and TPT layers, the heat flux toward the glass surface was larger than that toward the TPT layer.

The temperature distribution across the layers was simulated to evaluate PV module performance. The module equipped with a TPT layer required 1 h for the temperature to decrease from 57 to 45 °C, as shown in Fig. 17. However, the model without the TPT layer required only 10 min for the temperature to decrease from 57 to 33 °C, as shown in Fig. 17. The TPT layer is made of a thermal insulating material with a low thermal conductivity (0.414 W/m°K), which makes the process of heat removal from the back of the PV module difficult and energy-consuming. Thus, multiple TEC units are required to achieve proper cooling. However, this is not a cost effective solution. A streamline plot was thus used to predict the direction of airflow to obtain an optimal cooling effect, and the flow of air in and out of the module had a significant impact on the cooling effect. As shown in Fig. 17, the design without the TPT layer achieves a higher cooling effect than that with the TPT layer. This can be attributed to the well-streamlined distribution in the former model.

4. Conclusion

This study aimed to enhance the PV module performance by decreasing the module temperature while maintaining the lowest energy consumption of the cooling system. A dual active-cooling technique was tested by integrating a WF on the top surface of the PV glass and two TEC units on the rear side. FEM numerical analysis was used to simulate the physics of heat transfer and temperature distribution within the PV module layers. The fluid (water) velocity and airflow were determined to be the main components of the dual-cooling technique. Experiments were conducted to test the performance of the PV modules as a function of the cooling. Thermal analysis was performed using a thermal infrared

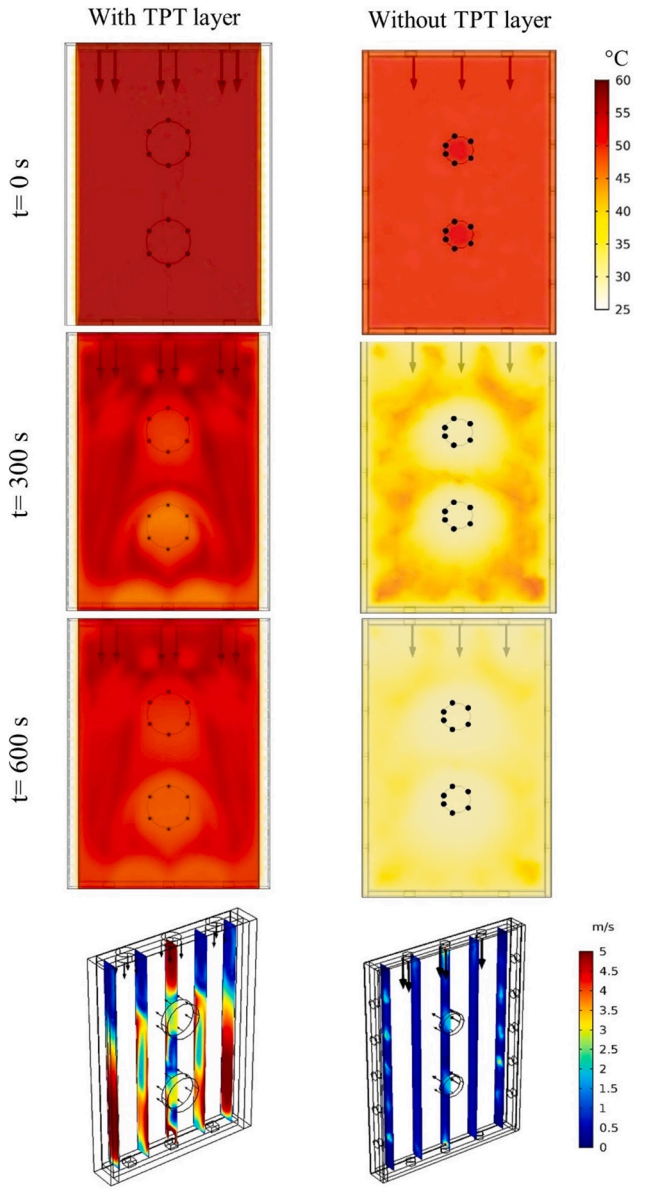


Fig. 17. Temperature distribution on the PV module surface and air velocity with and without the TPT layer at different times of 0, 300, 600 s; numeric modeling using FEM was used to simulate heat transfer and fluid flow owing to the TEC device cooling impact.

camera. The simulation results are consistent with the experimental data. When the solar density hitting the PV module surface increased, the output current also increased. The output voltage increased slightly at the same operating temperature. Similarly, the output power and energy from the PV modules increased with an increase in solar radiation density. A 21-°C temperature decrease was experienced in the PV module compared with the NCS, which caused an increase in efficiency of 11.23 %. Additionally, the output of the PV module improved by 9.23 % with the cooling system compared with that of the NCS. The maximum temperature was observed in the solar cell layer, which is the effective layer inside the PV module where radiation absorption occurs. The TPT layer was made from a material with low thermal conductivity (0.414 W/m K); therefore, it was difficult to remove heat from the backside of the PV module. Additional research should be conducted to solve the low thermal conductivity of the backside of the PV module to prevent the accumulation of heat inside the PV module layers. We suggest that changing the TPT layer with a high thermal conductivity material can enhance the natural cooling from the backside. In addition, a PVT collector based on a material with a high thermal conductivity will be investigated in future research. water-thin film cooling system (WFCS) led to better results than TEC, especially in the case of large-scale PV module systems, owing to the higher thermal conductivity of the water compared with the low thermal efficiency of the TEC units. Hence, we conclude that the WFCS is cleaner and more cost efficient than the TEC-based technique.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the Korea Institute of Planning and Evaluation for Technology in Food, Agriculture, Forestry, and Fisheries (IPET) through the Agricultural Energy Self-Sufficient Industrial Model Development Program, funded by Ministry of Agriculture, Food, and Rural Affairs (MAFRA; No. 321007-2).

References

Abdelrahman, M., Eliwa, A., Abdellatif, O.E., 2013. Experimental investigation of different cooling methods for photovoltaic module. In: 11th International Energy Conversion Engineering Conference July 14–17, San Jose, CA, 4096. https://doi.org/10.2514/6.2013-4096.

Aly, S.P., Barth, N., Figgis, B., Essadiqi, E., Faqir, M., Ennaoui, A., Ahzi, S., 2016. Mitigating the effect of heat and dust to enhance solar panels efficiency. In: 2016 International Renewable and Sustainable Energy Conference (IRSEC), 835-841. https://doi.org/10.1109/IRSEC.2016.7983870.

Aneli, S., Arena, R., Gagliano, A., 2021. Numerical Simulations of a PV Module with phase change material (PV-PCM) under Variable Weather Conditions. Int. J. Heat Technol. 39 (2), 643–652.

Borges Neto, M.R.B., Carvalho, P.C.M., Carioca, J.O.B., Canafístula, F.J.F., 2010. Biogas/photovoltaic hybrid power system for decentralized energy supply of rural areas. Energy Policy 38 (8), 4497–4506. https://doi.org/10.1016/j.enpol.2010.04.004.

Bouraiou, A., Hamouda, M., Chaker, A., Mostefaoui, M., Lachtar, S., Sadok, M., Boutasseta, N., Othmani, M., Issam, A., 2015. Analysis and evaluation of the impact of climatic conditions on the photovoltaic modules' performance in the desert environment. Energy Convers. Manage. 106, 1345–1355. https://doi.org/10.1016/j.enconman.2015.10.073.

Brahim, T., Jemni, A., 2017. Economical assessment and applications of photovoltaic/thermal hybrid solar technology: A review. Sol. Energy 153, 540–561. https://doi.org/10.1016/j.solener.2017.05.081.

Chandrasekar, M., Senthilkumar, T., 2021. Five decades of evolution of solar photovoltaic thermal (PVT) technology – A critical insight on review articles. J Clean Prod. 322, 128997. https://doi.org/10.1016/j.jclepro.2021.128997.

Da Silva, R.M., Fernandes, J.L.M., 2010. Hybrid photovoltaic/thermal (PV/T) solar systems simulation with Simulink/MATLAB. Sol. Energy 84 (12), 1985–1996. https://doi.org/10.1016/j.enconman.2015.10.073.

Dannemand, M., Perers, B., Furbo, S., 2019. Performance of a demonstration solar PVT assisted heat pump system with cold buffer storage and domestic hot water storage tanks. Energy Build. 188, 46–57. https://doi.org/10.1016/j.enbuild.2018.12.042.

Dimri, N., Tiwari, A., Tiwari, G.N., 2018. Effect of thermoelectric cooler (TEC) integrated at the base of opaque photovoltaic (PV) module to enhance an overall electrical efficiency. Sol. Energy 166, 159–170. https://doi.org/10.1016/j.solener.2018.03.030.

Enescu, D., Spertino, F., 2017. Applications of hybrid photovoltaic modules with thermoelectric cooling. Energy Proc. 111, 904–913. https://doi.org/10.1016/j.egypro.2017.03.253.

Ghadikolaei, S.S.C., 2021a. An enviroeconomic review of the solar PV cells cooling technology effect on the CO2 emission reduction. Sol. Energy 216, 468–492. https://doi.org/10.1016/j.solener.2021.01.016.

Ghadikolaei, S.S.C., 2021b. Solar photovoltaic cells performance improvement by cooling technology: An overall review. Int. J. Hydrog. Energy 46 (18), 10939–10972. https://doi.org/10.1016/j.ijhydene.2020.12.164.

Ghadikolaei, S.S., Yassari, M., Sadeghi, H., Hosseinzadeh, K., Ganji, D.D., 2017. Investigation on thermophysical properties of Tio2–Cu/H2O hybrid nanofluid transport dependent on shape factor in MHD stagnation point flow. Powder Technol. 322, 428–438. https://doi.org/10.1016/j.powtec.2017.09.006.

Ghadikolaei, S.S., Hosseinzadeh, K., Ganji, D.D., 2018a. Investigation on three dimensional squeezing flow of mixture base fluid (ethylene glycol-water) suspended by hybrid nanoparticle (Fe3O4-Ag) dependent on shape factor. J. Mol. Liq. 262, 376–388. https://doi.org/10.1016/j.molliq.2018.04.094.

Ghadikolaei, S.S., Hosseinzadeh, K., Ganji, D.D., 2018b. Numerical study on magnetohydrodynic CNTs-water nanofluids as a micropolar dusty fluid influenced by non-linear thermal radiation and joule heating effect. Powder Technol. 340, 389–399. https://doi.org/10.1016/j.powtec.2018.09.023.

Ghadikolaei, S.S., Hosseinzadeh, K., Hatami, M., Ganji, D.D., Armin, M., 2018c. Investigation for squeezing flow of ethylene glycol (C2H6O2) carbon nanotubes (CNTs) in rotating stretching channel with nonlinear thermal radiation. J. Mol. Liq. 263, 10–21. https://doi.org/10.1016/j.molliq.2018.04.141.

Hasanuzzaman, M., Rahim, N.A., Hosenuzzaman, M., Saidur, R., Mahbubul, I.M., Rashid, M.M., 2012. Energy savings in the combustion-based process heating in industrial sector. Renew. Sustain. Energy Rev. 16 (7), 4527–4536. https://doi.org/10.1016/j.rser.2012.05.027.

Hasanuzzaman, M., Malek, A.B.M.A., Islam, M.M., Pandey, A.K., Rahim, N.A., 2016. Global advancement of cooling technologies for PV systems: A review. Sol. Energy 137, 25–45. https://doi.org/10.1016/j.solener.2016.07.010.

Hindee, I., Lampongchat, G., Suwapaet, N., 2016. Performance study of solar panels with cooling systems at low ambient temperature. KKU Eng. J. 43, 47–51. https://doi.org/10.14456/kkuenj.2016.67.

Hosseini, R., Naghmeh, H., Khorasanizadeh, H., 2011. An experimental study of combining a photovoltaic system with a heating system. World Renewable Energy Congress 2011- Sweden 8-13 May, Linkoping. Sweden, 2993-3000. https://doi.org/10.3384/ecp110572993.

Joshi, A.S., Tiwari, A., 2007. Energy and exergy efficiencies of a hybrid photovoltaic–thermal (PV/T) air collector. Renew. Energy 32 (13), 2223–2241. https://doi.org/10.1016/j.renene.2006.11.013.

Kalaugher, L., 2014. Olive production: You win some, you lose some. Australian & New Zealand Olivegrower & Processor. Nation. J. Olive Indust. 2014 (94), 28–29. https://search.informit.org/doi/10.3316/informit.914292449837562.

Kane, A., Verma, V., Singh, B., 2017. Optimization of thermoelectric cooling technology for an active cooling of photovoltaic panel. Renew. Sustain. Energy Rev. 75, 1295–1305. https://doi.org/10.1016/j.rser.2016.11.114.

Khandelwal, S., Reddy, K.S., Murthy, S.S., 2007. Performance of contact and non-contact type hybrid photovoltaic-thermal (PV-T) collectors. Int. J. Low-Carbon Technol. 2 (4), 359–375.

Kotkondawar, A., Bhende, A., Khond, V., Rayalu, S., 2021. Performance evaluation of PV-TEC coupling device for power production with improved hybrid nanocarbon based thermal material interface. Energy Rep. 7, 6868–6875. https://doi.org/10.1016/j.egyr.2021.09.110.

Kumar, A., Baredar, P., Qureshi, U., 2015. Historical and recent development of photovoltaic thermal (PVT) technologies. Renew. Sustain. Energy Rev. 42, 1428–1436. https://doi.org/10.1016/j.rser.2014.11.044.

Manning, H.E., Chang, J.S., Fang, H., Hammond, G., Yang, Y., Lee, S.Y., 2019. BMC Energy: a home for all energy and fuels research. BMC Energy 1 (1), 1–4. https://doi.org/10.1186/s42500-019-0003-8.

Moshfegh, H., Eslami, M., Hosseini, A., 2018. Thermoelectric cooling of a photovoltaic panel. In: Nižetić, S., Papadopoulos, A. (Eds.), The Role of Exergy in Energy and the Environment. Springer, pp. 625–634. https://doi.org/10.1007/978-3-319-89845-2_44.

Najafi, H., Woodbury, K.A., 2013. Optimization of a cooling system based on Peltier effect for photovoltaic cells. Solar Energy 91, 152-160. https://doi.org/10.1016/j.solener.2013.01.026.

Rahman, M.M., Hasanuzzaman, M., Rahim, N.A., 2015. Effects of various parameters on PV-module power and efficiency. Energy Convers. Manage. 103, 348–358. https://doi.org/10.1016/j.enconman.2015.06.067.

Razak, A., Irwan, Y., Leow, W., Irwanto, M., Safwati, I., Zhafarina, M., 2016. Investigation of the effect temperature on photovoltaic (PV) panel output performance. Int. J. Adv. Sci. Eng. Inform. Technol. 6 (5), 682–688. https://doi.org/10.18517/ijaseit.6.5.938.

Renno, C., 2014. Optimization of a concentrating photovoltaic thermal (CPV/T) system used for a domestic application. Appl. Therm. Eng. 67 (1), 396–408. https://doi.org/10.1016/j.applthermaleng.2014.03.026.

Siecker, J., Kusakana, K., Numbi, B.P., 2017. A review of solar photovoltaic systems cooling technologies. Renew. Sustain. Energy Rev. 79, 192–203. https://doi.org/10.1016/j.rser.2017.05.053.

Singh, D., Chaubey, H., Parvez, Y., Monga, A., Srivastava, S., 2022. Performance improvement of solar PV module through hybrid cooling system with thermoelectric coolers and phase change material. Solar Energy 241, 538–552. https://doi.org/10.1016/j.solener.2022.06.028.

Teo, H.G., Lee, P.S., Hawlader, M.N.A., 2012. An active cooling system for photovoltaic modules. Appl. Energy 90 (1), 309–315. https://doi.org/10.1016/j.apenergy.2011.01.017.

Terashima, K., Sato, H., Ikaga, T., 2020. Development of an environmentally friendly PV/T solar panel. Sol. Energy 199, 510–520. https://doi.org/10.1016/j.solener.2020.02.051.

Zubeer, S.A., Ali, O.M., 2022. Experimental and numerical study of low concentration and water-cooling effect on PV module performance. Case Stud. Therm. Eng. 34, 102007. https://doi.org/10.1016/j.csite.2022.102007.